Introduction to Deep Equilibrium Nets (DEQNs)

A Mini-Course — March 2026

Description

This self-contained mini-course introduces **Deep Equilibrium Nets (DEQNs)**, a method that uses deep neural networks to solve high-dimensional dynamic stochastic economic models. DEQNs replace traditional grid-based collocation with neural network training, where the loss function is derived directly from the economic equilibrium conditions — no labeled data is needed.

The course covers the necessary ML/DL foundations, develops the DEQN methodology from scratch, applies it to the Brock–Mirman benchmark (which has an analytical solution), scales it up to the multi-country International Real Business Cycle (IRBC) model, and introduces practical tools such as Neural Architecture Search and loss normalization (ReLoBRaLo).

Prerequisites

- Basic knowledge of macroeconomics (Euler equations, dynamic programming)
- Familiarity with Python and Jupyter notebooks
- Linear algebra and calculus fundamentals

Software Requirements

Package	Version
Python	>= 3.9
NumPy	>= 1.24
SciPy	>= 1.10
Matplotlib	>= 3.7
TensorFlow	>= 2.15
TensorFlow Probability	>= 0.23
PyTorch	>= 2.0

Install all dependencies via:

```
pip install -r requirements.txt
```

Course Content

The slide deck (`slides/DEQN_MiniCourse.tex`) is organized as follows:

Part	Topic	Content
I	ML/DL Foundations	DNNs as function approximators, activation functions, SGD, loss functions
II	Deep Equilibrium Nets	Motivation, curse of dimensionality, DEQN loss function, training algorithm

Part	Topic	Content
III	Brock–Mirman Benchmark	Analytical test case, DEQN implementation, parallelization & scalability
IV	Practical Tools	Neural Architecture Search (NAS) and ReLoBRaLo loss normalization
V	Scaling Up — The IRBC Model	Multi-country model, equilibrium system, high-dimensional DEQNs
VI	Hands-on Exercises	Notebook overview and suggested readings

Code Notebooks

Notebook	Description
`code/01_Brock_Mirman_1972_DEQN.ipynb`	Deterministic Brock–Mirman model with analytical benchmark
`code/02_Brock_Mirman_Uncertainty_DEQN.ipynb`	Stochastic Brock–Mirman with TFP shocks
`code/03_DEQN_Exercises_Blancs.ipynb`	Exercise problems (blanks to fill in)
`code/03_DEQN_Exercises_Solutions.ipynb`	Exercise solutions
`code/04_IRBC_DEQN.ipynb`	Full IRBC model: multi-country DEQN with adjustment costs and irreversibility

Readings

Paper	File
Azinovic, Gaegauf & Scheidegger (2022). *Deep Equilibrium Nets.* International Economic Review 63(4), 1471–1525.	`readings/Azinovic_Gaegauf_Scheidegger_2022_DEQN.pdf`
Fernández-Villaverde, Nuño & Perla (2024). *Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning.* NBER Working Paper 33117.	`readings/taming.pdf`
Chen, Didisheim & Scheidegger (2026). *Deep Surrogates for Finance: With an Application to Option Pricing.* Journal of Financial Economics 177, 104222.	`readings/DeepSurrogates_JFE.pdf`
Friedl, Kübler, Scheidegger & Usui (2023). *Deep Uncertainty Quantification: With an Application to Integrated Assessment Models.* Working paper.	`readings/DeepUQ_with_an_application_to_IAM.pdf`

References

If you use materials from this course, please cite:

```
@article{azinovic2022deep,
  title={Deep Equilibrium Nets},
  author={Azinovic, Marina and Gaegauf, Luca and Scheidegger, Simon},
  journal={International Economic Review},
  volume={63},
  number={4},
  pages={1471--1525},
  year={2022},
  doi={10.1111/iere.12575}
}
```

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